Validity-Assured Regeneration of Parametric Solid Modeling : Supplementary Material

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For simplicity, we ignore $|\cdot|$ when there is no ambiguity. As an example, equation

$$\partial\widehat{C} = \left|\left(F_{\widehat{A}} \setminus \mathrm{gen}_A^-\right) \cup \left(F_{\widehat{B}} \setminus \mathrm{gen}_B^-\right) \cup \mathrm{gen}^+\right|. \tag{1}$$

can be written in simpler form:

$$\partial\widehat{C} = \left(F_{\widehat{A}} \setminus \mathrm{gen}_A^-\right) \cup \left(F_{\widehat{B}} \setminus \mathrm{gen}_B^-\right) \cup \mathrm{gen}^+. \tag{2}$$

In the following discussion, the symbol "−" is overloaded: depending on the operated terms, it denotes either point-set subtraction or set subtraction.

As stated in the main text, we consider only valid solid B-reps $A, \widehat{A}, B, \widehat{B}$ with manifold boundaries.

For a point set S, we denote its interior points by S°, and its complement by S^c in the following text.

The notation $d(x, y)$ denotes the Euclidean distance between two points x and y.

The notation $d(S, T)$ denotes the distance between two point sets S and T.

The ϵ-neighborhood of a point P, denoted by $U(P, \epsilon)$, is defined as:

$$d(S,T) \triangleq inf\{d(x,y) \mid x \in S, y \in T\},$$
$$U(P,\epsilon) \triangleq \{x \in R^3 \mid d(x,P) < \epsilon\}.$$

1 Completion of Proof in Main Text

In this section, we adapt *Criterion 1-2* and *Assumption 1-2* to the Boolean union case, based on which we complete the proof presented in the main text.

1.1 Assumptions and Criteria

Assumption 1	$\partial A,\ \partial\widehat{A},\ \partial B,\ \partial\widehat{B}$ are connected.
Assumption 2	$\partial\widehat{A} \cap \partial\widehat{B} \neq \emptyset$.
Criterion 1	$\mathrm{gen}_A^- \subseteq F_{\widehat{A}}, \quad \mathrm{gen}_B^- \subseteq F_{\widehat{B}}$.
Criterion 2	$\forall f \in \mathrm{newf}_A, \quad f \cap \partial\widehat{B} = \emptyset,$ $\forall f \in \mathrm{newf}_B, \quad f \cap \partial\widehat{A} = \emptyset.$

1.2 Lemmas

Proposition 1.2.1 states that any point P on intersection curve $\partial A \cap \partial B$ has a neighborhood $U(P, \epsilon_P)$, in which the solid regions remain invariant during regeneration (Fig. 1).

Proposition 1.2.1.

$$P \in \partial A \cap \partial B \Longrightarrow \exists \epsilon_P > 0,\ s.t.\ \begin{aligned} U(P,\epsilon_P) \cap \widehat{A} &= U(P,\epsilon_P) \cap A. \\ U(P,\epsilon_P) \cap \widehat{B} &= U(P,\epsilon_P) \cap B. \\ U(P,\epsilon_P) \cap \widehat{C} &= U(P,\epsilon_P) \cap C. \end{aligned}$$

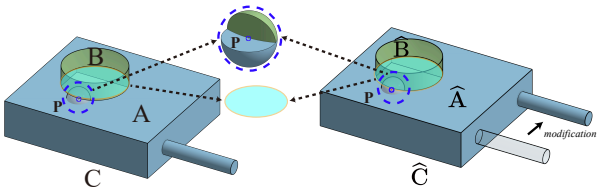


Fig. 1. Figure of *Prop* 1.2.1 and *Lem* 1.2.1. Under conditions in Sec. 1.1, some 3-dimensional neighborhood of P (highlighted in blue dashed line) and the intersection $\partial A \cap \partial B$ of ∂A(the green surface) and ∂B (the light blue surface) remain the same after the modification.

Proof. Let f be any face in F_A that contains P. Then

$$P \in f \cap B \Longrightarrow f \cap B \neq \emptyset \Longrightarrow f \in gen_A^- \overset{Criterion\ 1}{\Longrightarrow} f \in F_{\widehat{A}}. \tag{3}$$

Hence, $P \in f \in F_A$ implies $f \in F_{\widehat{A}}$, and therefore $P \in \partial\widehat{A}$.

Similarly, we obtain $P \in \partial\widehat{B}$. Now consider $P \in g \in F_{\widehat{A}}$. If $g \in newf_A$, then we can derive

$$g \in newf_A \overset{P \in g \cap \partial\widehat{B}}{\Longrightarrow} g \cap \partial\widehat{B} \neq \emptyset, \tag{4}$$

which contradicts *Criterion 2*. Therefore, we have

$$P \in g \in F_{\widehat{A}} \Longrightarrow \begin{matrix} g \in F_{\widehat{A}} \\ g \notin newf_A \end{matrix} \Longrightarrow g \in F_A. \tag{5}$$

Define $O_A \triangleq \{f \in F_A \cup F_{\widehat{A}} \mid P \notin f\}$, then O_A is a finite face set.

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From definition we know $\forall f_A \in O_A,\ d(P, f_A) > 0$. Since O_A is finite, it follows that $d(P, O_A) > 0$. Let $\epsilon_A = \frac{1}{2}d(P, O_A)$, then F_A and $F_{\widehat{A}}$ are totally equal in $U(P, \epsilon_A)$, which implies

$$U(P, \epsilon_A) \cap \widehat{A} = U(P, \epsilon_A) \cap A. \tag{6}$$

Similarly, $\exists \epsilon_B > 0\ s.t.\ U(P, \epsilon_B) \cap \widehat{B} = U(P, \epsilon_B) \cap B$.

Define $\epsilon_P = min(\epsilon_A, \epsilon_B)$. Then the conclusion follows from

$$C = A \cup B, \quad \widehat{C} = \widehat{A} \cup \widehat{B}.$$

□

Lemma 1.2.1.

$$\partial A \cap \partial B = \partial\widehat{A} \cap \partial\widehat{B}. \tag{7}$$

Proof. By *Proposition 1.2.1*, we directly have

$$\partial A \cap \partial B \subseteq \partial\widehat{A} \cap \partial\widehat{B}. \tag{8}$$

It remains to prove $\partial\widehat{A} \cap \partial\widehat{B} \subseteq \partial A \cap \partial B$. Suppose that

$$P \in \partial\widehat{A} \cap \partial\widehat{B}, P \notin \partial A \cap \partial B, \tag{9}$$

By symmetry, we may assume $P \notin \partial A$. Then

$$\begin{aligned} P &\in \partial\widehat{A} - \partial A \subseteq new f_A, \\ P &\in \partial\widehat{A} \cap \partial\widehat{B} \subseteq \partial\widehat{B}, \end{aligned} \tag{10}$$

which contradicts *Criterion 2*. Therefore,

$$P \in \partial\widehat{A} \cap \partial\widehat{B} \Longrightarrow P \in \partial A \cap \partial B.$$

This proves the lemma (Fig. 1).

□

Lemma 1.2.2.

$$\partial\widehat{A} \cap \partial\widehat{B} \cap \partial\widehat{C} = \partial A \cap \partial B \cap \partial C.$$

Proof. Directly follows from *Proposition 1.2.1* and *Lemma 1.2.1*.

□

Before proving *Lemma 1.2.3* and *Lemma 3* in the main text, we first introduce *Proposition 1.2.2*, which describes the local behavior of two manifold surfaces in a neighborhood of their intersection curve.

Proposition 1.2.2.

$$\begin{aligned} &Let && M, N \ \ be\ 2-dimensional\ manifolds, \\ &then && M \cap N \cap \overline{M - N} \subseteq \overline{N - M}. \end{aligned}$$

Proof. Assume, for contradiction, that there exists a point $P \in M \cap N \cap \overline{M - N}$ such that $P \notin \overline{N - M}$.

Since $P \notin \overline{N - M}$, $\exists$ open neighborhood U of P such that

$$N \cap U \subseteq M \cap U.$$

By the manifold condition, there exist open neighborhoods V_1, V_2 of P such that

$$\exists \text{ homeomorphism } \sigma_1 : \mathbb{R}^2 \to V_1 \cap M, \quad \sigma_2 : \mathbb{R}^2 \to V_2 \cap N.$$

$$\begin{array}{ccc} N \cap W & \xleftarrow{\sigma_1} & \mathbb{R}^2 \\ \downarrow i & & \downarrow \phi \\ M \cap W & \xleftarrow{\sigma_2} & \mathbb{R}^2 \end{array}$$

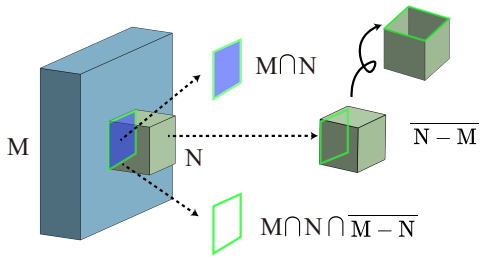


Fig. 2. Figure of *Prop* 1.2.2. Here M (the blue box) and N (the green box) are 2-dimensional manifolds placed next to each other, and $M \cap N \cap \overline{M - N}$ (the light green loop) $\subseteq \overline{N - M}$ (the green box with an open face removed).

Define $\phi \triangleq \sigma_2^{-1} \circ i \circ \sigma_1$, where i denotes the inclusion map, then ϕ is a continuous injection.

By the invariance of domain theorem, $\phi(\mathbb{R}^2)$ is open in $\mathbb{R}^2$.

Therefore, $i \circ \sigma_1(\mathbb{R}^2) = \sigma_2(\phi(\mathbb{R}^2))$ is open in $M \cap W$ as well.

Since $P \in N \cap W = i \circ \sigma_1(\mathbb{R}^2)$, this implies

$$P \notin \overline{(M \cap W) - (N \cap W)} = \overline{W \cap (M - N)}.$$

Since W is an open neighborhood of P, we have $P \notin \overline{M - N}$, which is a contradiction.

Thus, for every $P \in M \cap N \cap \overline{M - N}$, it follows that $P \in \overline{N - M}$.

□

Lemma 1.2.3.

$$\partial\widehat{A} \cap \widehat{B}^\circ = \partial A \cap B^\circ.$$

Proof. For $\forall P \in \partial\widehat{A} \cap \widehat{B}^\circ$, we know $\exists X_0 \in \partial\widehat{A} \cap \partial\widehat{B}$ from *Assumption 2*. Together with *Assumption 1* we know:

$$\exists \gamma : [0, 1] \to \partial\widehat{A}, \quad \begin{aligned} \gamma(0) &= P, \\ \gamma(1) &= X_0. \end{aligned}$$

Moreover, we may assume that $\gamma([0, 1)) \cap \partial\widehat{B} = \emptyset$. Otherwise, we can let X^* be the first intersection point between γ and $\partial\widehat{B}$, and let $\widehat{\gamma}$ be the path from P to X^*. Then we have

$$\gamma(0) \in \widehat{B}^\circ \ \Rightarrow\ \gamma([0, 1)) \subseteq \widehat{B}^\circ.$$

From *Proposition 1.2.1*, we know that $\exists \epsilon > 0,\ s.t.\ U\ open$, and

$$\gamma((1 - \epsilon, 1)) \subseteq U, \quad \begin{aligned} A \cap U &= \widehat{A} \cap U, \\ B \cap U &= \widehat{B} \cap U, \\ C \cap U &= \widehat{C} \cap U. \end{aligned}$$

Therefore,

$$\begin{aligned} \gamma((1 - \epsilon, 1)) \subseteq \partial\widehat{A} &\Rightarrow \gamma((1 - \epsilon, 1)) \subseteq \partial A, \\ \gamma((1 - \epsilon, 1)) \subseteq \widehat{B}^\circ &\Rightarrow \gamma((1 - \epsilon, 1)) \subseteq B^\circ. \end{aligned}$$

Define

$$S_1 = \{\, t \in [0,1) \mid \gamma(t) \notin \partial A \,\},$$
$$S_2 = \{\, t \in [0,1) \mid \gamma(t) \notin B^\circ \,\}.$$

If $S_1 \cup S_2 \neq \emptyset$, let $t_0 = \sup(S_1 \cup S_2) \leq 1-\epsilon$:

(1) $t_0 \in S_1 (Fig.\ 3)$:

By the definition of S_1, we have

$$\gamma(t_0) \in \partial\widehat{A} \cap \partial A \cap \overline{\partial\widehat{A} - \partial A}.$$

Then, by *Proposition 1.2.2*, we obtain

$$\gamma(t_0) \in \overline{\partial A - \partial\widehat{A}}.$$

Consequently, $\exists \alpha \in F_A,\ s.t.\ \gamma(t_0) \in \alpha, \alpha \notin F_{\widehat{A}}$. Moreover,

$$\sup S_2 \leq t_0 \Rightarrow \gamma((t_0,1)) \subseteq B^\circ \Rightarrow \gamma(t_0) \in B.$$

Now,

$$\gamma(t_0) \in \alpha \cap B \Longrightarrow \alpha \in \mathrm{gen}_A^- \overset{Criterion\ 1}{\Longrightarrow} \alpha \in F_{\widehat{A}},$$

which yields a contradiction.

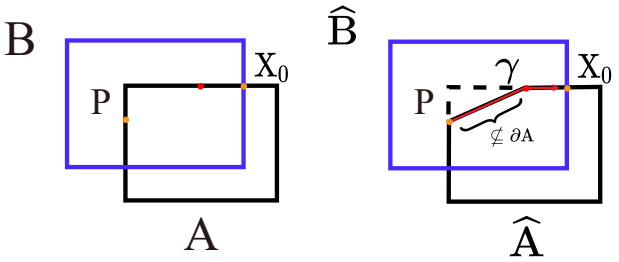


Fig. 3. A 2-dimensional example of *Lem* 1.2.3 failure case 1, where part of γ from P is away from ∂A.

(2) $t_0 \in S_2 (Fig.\ 4)$:

In this case, by the definition of S_2, we have $\gamma(t_0) \in \partial B$. Moreover,

$$\sup S_1 \leq t_0 \Rightarrow \gamma((t_0,1)) \subseteq \partial A \Rightarrow \gamma(t_0) \in \partial A.$$

Therefore,

$$\gamma(t_0) \in \partial A \cap \partial B = \partial\widehat{A} \cap \partial\widehat{B}.$$

by *Lemma 1.2.1*, which contradicts our previous assumption that $\gamma([0,1)) \cap \partial\widehat{B} = \emptyset$.

From the above two cases, we conclude that $S_1 \cup S_2 = \emptyset$. Hence, $P \in \partial A \cap B^\circ$, and therefore $\partial\widehat{A} \cap \widehat{B}^\circ \subseteq \partial A \cap B^\circ$.

For $\forall Q \in \partial A \cap B^\circ$, let $Q \in \alpha_0 \in F_A$. Then $\alpha_0 \in \mathrm{gen}_A^- \subseteq \partial\widehat{A}$ by *Criterion 1*, and hence $Q \in \partial\widehat{A}$.

By *Assumption 2* and *Lemma 1.2.1*, we know that $\exists X_0 \in \partial A \cap \partial B$. Together with *Assumption 1*, we know:

$$\exists \gamma : [0,1] \to \partial A, \quad \begin{matrix} \gamma(0) = Q, \\ \gamma(1) = X_0. \end{matrix}$$

Moreover, we may assume that $\gamma([0,1)) \cap \partial B = \emptyset$. Otherwise, we can let X^* be the first intersection point between γ and ∂B, and let $\widehat{\gamma}$ be the path from Q to X^*. Then we have

$$\gamma(0) \in B^\circ \ \Rightarrow\ \gamma([0,1)) \subseteq B^\circ.$$

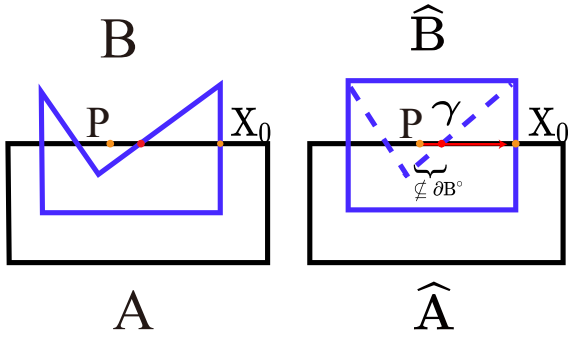


Fig. 4. A 2-dimensional example of *Lem* 1.2.3 failure case 2, where part of γ from P is away from B°.

From *Proposition 1.2.1*, we know that $\exists \epsilon > 0,\ s.t.\ U\ open$, and

$$\gamma((1-\epsilon,1)) \subseteq U, \quad \begin{matrix} A \cap U = \widehat{A} \cap U, \\ B \cap U = \widehat{B} \cap U, \\ C \cap U = \widehat{C} \cap U. \end{matrix}$$

Therefore,

$$\gamma((1-\epsilon,1)) \subseteq \partial A \Rightarrow \gamma((1-\epsilon,1)) \subseteq \partial\widehat{A},$$
$$\gamma((1-\epsilon,1)) \subseteq B^\circ \Rightarrow \gamma((1-\epsilon,1)) \subseteq \widehat{B}^\circ.$$

Define

$$S_1 = \{\, t \in [0,1) \mid \gamma(t) \notin \partial\widehat{A} \,\},$$
$$S_2 = \{\, t \in [0,1) \mid \gamma(t) \notin \widehat{B}^\circ \,\}.$$

If $S_1 \cup S_2 \neq \emptyset$, let $t_0 = \sup(S_1 \cup S_2) \leq 1-\epsilon$:

(1) $t_0 \in S_1$:

We know that $\exists \alpha \in F_A,\ s.t.\ \gamma(t_0) \in \alpha, \alpha \notin F_{\widehat{A}}$. Hence,

$$\gamma(t_0) \in \alpha \cap B \Longrightarrow \alpha \in \mathrm{gen}_A^- \overset{Criterion\ 1}{\Longrightarrow} \alpha \in F_{\widehat{A}},$$

which yields a contradiction.

(2) $t_0 \in S_2$:

In this case, by the definition of S_2, we have $\gamma(t_0) \in \partial\widehat{B}$. Moreover,

$$\sup S_1 \leq t_0 \Rightarrow \gamma((t_0,1)) \subseteq \partial\widehat{A} \Rightarrow \gamma(t_0) \in \partial\widehat{A}.$$

Therefore,

$$\gamma(t_0) \in \partial\widehat{A} \cap \partial\widehat{B} = \partial A \cap \partial B.$$

by *Lemma 1.2.1*, which contradicts our previous assumption that $\gamma([0,1)) \cap \partial B = \emptyset$.

From the above two cases, we conclude that $S_1 \cup S_2 = \emptyset$. Hence, $Q \in \partial\widehat{A} \cap \widehat{B}^\circ$, and therefore $\partial A \cap B^\circ \subseteq \partial\widehat{A} \cap \widehat{B}^\circ$.

□

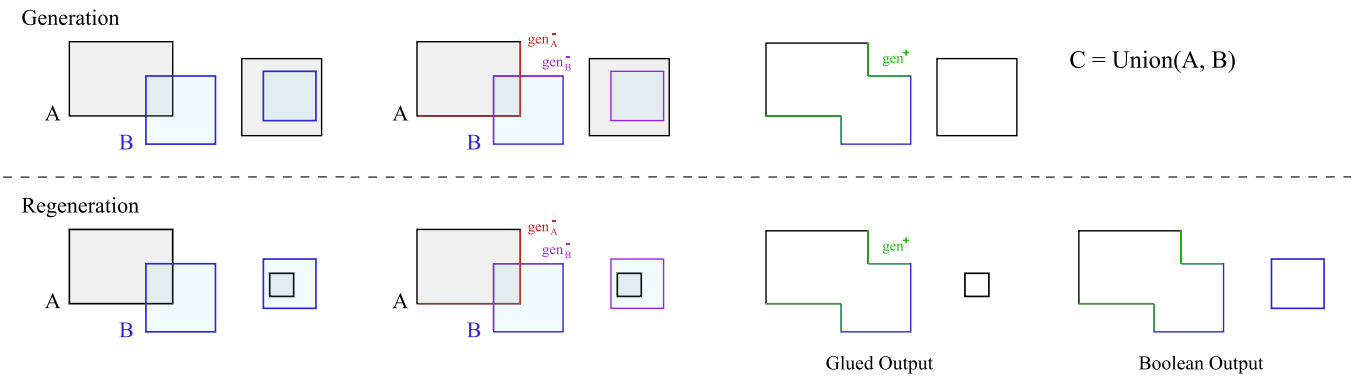


Fig. 5. Failure case when assumptions are not satisfied

Proposition 1.2.4.

$$gen^+ \cap \partial A \subseteq gen_A^-. \tag{11}$$

Proof. We prove this proposition by contradiction. Suppose that $gen^+ \cap \partial A \nsubseteq gen_A^-$. Then there $\exists P \in gen^+ \cap \partial A, P \notin gen_A^-$.

Assume that point P lies on faces α_i $(1 \le i \le m)$ in F_A and faces β_j $(1 \le j \le n)$ in F_B, and $m, n \ge 0$.

Since $P \notin gen_A^-$, we obtain

$$\alpha_i \cap B = \emptyset, \qquad \alpha_i \in F_C, \qquad \alpha_i \notin gen^+.$$

Since $P \in gen^+$, there exists a face γ such that $P \in \gamma, \gamma \in gen^+$. By the definitions of α_i and β_j, there exists an open neighborhood U of P such that

$$U \cap \partial A = U \cap (\cup_{i=1}^m \alpha_i)\ , \quad U \cap \partial B = U \cap (\cup_{j=1}^n \beta_j),$$

Then we have

$$U \cap \gamma \subseteq U \cap \partial C \subseteq U \cap (\partial A \cup \partial B) \subseteq U \cap ((\cup_{i=1}^m \alpha_i) \cup (\cup_{j=1}^n \beta_j).$$

Let m_2 denote the two-dimensional measure induced on ∂C. Then

$$\begin{aligned} 0 <& m_2(U \cap \gamma) \\ \le& \Sigma_{i=1}^m m_2(U \cap \gamma \cap \alpha_i) + \Sigma_{j=1}^n m_2(U \cap \gamma \cap \beta_j) \\ \le& \Sigma_{i=1}^m m_2(\gamma \cap \alpha_i) + \Sigma_{j=1}^n m_2(\gamma \cap \beta_j). \end{aligned}$$

Since $\alpha_i \notin gen^+, \gamma \in gen^+$, we have $\alpha_i \neq \gamma, \forall i$. Together with $\alpha_i, \gamma \in F_C$, this implies that α_i and γ are distinct faces in the valid B-rep model C. Hence, their intersection is at most one-dimensional, and therefore $m_2(\alpha_i \cap \gamma) = 0$. As a consequence,

$$\begin{aligned} 0 <& \Sigma_{i=1}^m m_2(\gamma \cap \alpha_i) + \Sigma_{j=1}^n m_2(\gamma \cap \beta_j) \\ =& \Sigma_{j=1}^n m_2(\gamma \cap \beta_j). \end{aligned}$$

Therefore, we must have $n > 0$, which implies $P \in \partial B$. Hence,

$$P \in \partial A \cap \partial B \Rightarrow P \in gen_A^-.$$

which yields a contradiction.

□

1.3 $\text{RHS}_A \subseteq \text{LHS}$ in Main Theorem

We break down the proof of this theorem into the following two parts (1.3.1 and 1.3.2) :

1.3.1 $F_{\widehat{A}} - gen_A^- \subseteq LHS$.

Proof. By the definition of gen_A^-, we have

$$\partial A \cap B \subseteq gen_A^-.$$

Therefore,

$$\begin{aligned} F_{\widehat{A}} - gen_A^- \subseteq& F_{\widehat{A}} - (\partial A \cap B) \\ =& F_{\widehat{A}} - (\partial \widehat{A} \cap \widehat{B}) \qquad (\textit{Lemma 1 and Lemma 3}) \\ =& \partial \widehat{A} \cap \widehat{B}^c \subseteq \partial \widehat{C} = LHS. \end{aligned}$$

□

1.3.2 $gen^+ \cap \partial A \subseteq LHS$.

Proof.

$$\begin{aligned} gen^+ \cap \partial A \subseteq& gen_A^- \qquad (\textit{Lemma 4}) \\ \subseteq& F_{\widehat{A}} = \partial \widehat{A}. \qquad (\textit{Criterion 1}) \end{aligned}$$

This gives

$$gen^+ \cap \partial A = gen^+ \cap \partial A \cap \partial \widehat{A}.$$

Therefore, the conclusion holds from the decomposition of $gen^+ \cap \partial A \cap \partial \widehat{A}$ with respect to $\widehat{B}^\circ$, $\partial \widehat{B}$, and $\widehat{B}^c$:

$$\begin{aligned} gen^+ \cap \partial A \cap \partial \widehat{A} \cap \widehat{B}^\circ =& gen^+ \cap \partial A \cap \partial A \cap B^\circ \qquad (\textit{Lemma 3}) \\ \subseteq& \partial C \cap \partial A \cap B^\circ = \emptyset, \\ gen^+ \cap \partial A \cap \partial \widehat{A} \cap \partial \widehat{B} \subseteq& \partial C \cap \partial A \cap \partial \widehat{A} \cap \partial \widehat{B} \\ =& \partial C \cap \partial A \cap \partial A \cap \partial B \qquad (\textit{Lemma 1}) \\ =& \partial \widehat{C} \cap \partial \widehat{A} \cap \partial \widehat{B} \qquad (\textit{Lemma 2}) \\ \subseteq& \partial \widehat{C} = \text{LHS}, \end{aligned}$$

$$gen^+ \cap \partial A \cap \partial \widehat{A} \cap \widehat{B}^c \subseteq \partial \widehat{A} \cap \widehat{B}^c \subseteq \partial \widehat{C} = \text{LHS}.$$

□

2 Notations and Definitions for Generic Cases

Our generic theory can:

(1) Handle cases that violate the assumptions in the main text.
(2) Support arbitrary Boolean operations.
(3) Consider gluing-based regeneration at the loop-level.

In order to describe the full theory clearly, we first introduce the necessary notations and algorithms in Section 2, and then prove the validity in Section 3.

2.1 Intersecting and Isolated Components

We divide the connected components of $\partial\widehat{A}$ into two types according to whether they intersect $\partial\widehat{B}$:

$$Int_A \triangleq \{ P \in \partial A \mid \exists \gamma : [0,1] \to \partial A,\ s.t.\ \gamma(0) = P, \gamma(1) \in \partial A \cap \partial B \},$$
$$Iso_A \triangleq \{ P \in \partial A \mid \nexists \gamma : [0,1] \to \partial A,\ s.t.\ \gamma(0) = P, \gamma(1) \in \partial A \cap \partial B \},$$
$$Int_{\widehat{A}} \triangleq \{ P \in \partial\widehat{A} \mid \exists \gamma : [0,1] \to \partial\widehat{A},\ s.t.\ \gamma(0) = P, \gamma(1) \in \partial\widehat{A} \cap \partial\widehat{B} \},$$
$$Iso_{\widehat{A}} \triangleq \{ P \in \partial\widehat{A} \mid \nexists \gamma : [0,1] \to \partial\widehat{A},\ s.t.\ \gamma(0) = P, \gamma(1) \in \partial\widehat{A} \cap \partial\widehat{B} \}.$$

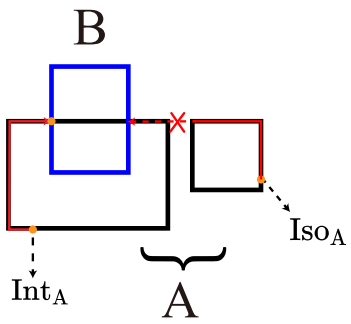


Fig. 6. A 2-dimensional example of Int_A and Iso_A

An example of Int_A and Iso_A is provided in Fig. 6. Similarly, Int_B, Iso_B, $Int_{\widehat{B}}$, $Iso_{\widehat{B}}$ are defined.

Without the assumptions made in the main text, isolated components may lead to incorrect results (Fig. 5). However, we can modify our algorithm by treating intersection components and isolated components separately, thereby obtaining the correct result.

Let ∂M_i be the ith connected component of ∂M, where $M \in \{A, B, \widehat{A}, \widehat{B}\}$. Let σ denote the Boolean operation; that is,

$$\sigma(A,B) \in \{A \cup B, A \cap B, A - B, B - A\}.$$

Proposition 2.1.1.

$$\partial\widehat{A}_i \subseteq Iso_{\widehat{A}} \ \Rightarrow\ \partial\widehat{A}_i \cap \partial\sigma(\widehat{A},\widehat{B}) \in \{\partial\widehat{A}_i, \emptyset\} \tag{12}$$

Proof. Let point $P \in \partial\widehat{A}_i$. By the definition of $Iso_{\widehat{A}}$, we have $P \notin \partial\widehat{B}$. Hence, either $P \in \widehat{B}^\circ$ or $P \in \widehat{B}^c$.

If $P \in \widehat{B}^\circ$, we claim that $\partial\widehat{A}_i \subseteq \widehat{B}^\circ$. Otherwise, there exists $Q \in \partial\widehat{A}_i$ such that $Q \in \widehat{B}^c$. Since $\partial\widehat{A}_i$ is a connected component, we can construct a path $\widehat{\gamma} \subseteq \partial\widehat{A}_i$ from P to Q. Then $\widehat{\gamma}$ intersects $\partial\widehat{B}$, which contradicts $\partial\widehat{A}_i \subseteq Iso_{\widehat{A}}$.

If $P \in \widehat{B}^c$, we can similarly prove that $\partial\widehat{A}_i \subseteq \widehat{B}^c$. Therefore, either $\partial\widehat{A}_i \subseteq \widehat{B}^\circ$ or $\partial\widehat{A}_i \subseteq \widehat{B}^c$. The result then follows by enumerating all Boolean operations as follows:

$$\partial\widehat{A}_i \cap \partial(\widehat{A} \cup \widehat{B}) = \begin{cases} \emptyset & (\partial\widehat{A}_i \subseteq \widehat{B}^\circ), \\ \partial\widehat{A}_i & (\partial\widehat{A}_i \subseteq \widehat{B}^c), \end{cases}$$
$$\partial\widehat{A}_i \cap \partial(\widehat{A} \cap \widehat{B}) = \begin{cases} \partial\widehat{A}_i & (\partial\widehat{A}_i \subseteq \widehat{B}^\circ), \\ \emptyset & (\partial\widehat{A}_i \subseteq \widehat{B}^c), \end{cases}$$
$$\partial\widehat{A}_i \cap \partial(\widehat{A} - \widehat{B}) = \begin{cases} \emptyset & (\partial\widehat{A}_i \subseteq \widehat{B}^\circ), \\ \partial\widehat{A}_i & (\partial\widehat{A}_i \subseteq \widehat{B}^c), \end{cases}$$
$$\partial\widehat{A}_i \cap \partial(\widehat{B} - \widehat{A}) = \begin{cases} \partial\widehat{A}_i & (\partial\widehat{A}_i \subseteq \widehat{B}^\circ), \\ \emptyset & (\partial\widehat{A}_i \subseteq \widehat{B}^c). \end{cases}$$

□

2.2 Regions for Arbitrary Boolean

We start this subsection by studying the behavior of isolated components during regeneration.

By *Proposition 2.1.1*, an isolated component $\partial\widehat{A}_i$ either entirely appears in or entirely disappears from the Boolean result $\sigma(\widehat{A},\widehat{B})$. To construct the correct result by gluing, we need to determine which case occurs. This can be done by performing a point-in-solid test for a point $P \in \partial\widehat{A}_i$ with respect to $\widehat{B}$.

To describe the construction of Boolean result from isolated components, we introduce *Definition 2.2.1*, and write down the way of validity-guaranteed construction of isolated components explicitly in *Proposition 2.2.2* and *Algorithm* 1.

Definition 2.2.1. For a Boolean operation σ, where

$$\sigma(A,B) \in \{A \cup B,\ A \cap B,\ A - B,\ B - A\},$$

we define

$$K_\sigma(B) \triangleq \begin{cases} B^\circ, & \sigma(A,B) \in \{A \cup B,\ A - B\}, \\ B^c, & \sigma(A,B) \in \{A \cap B,\ B - A\}, \end{cases}$$
$$L_\sigma(B) \triangleq \overline{K_\sigma(B)}^c.$$

For simplicity, we use notation $K(B) = K_\sigma(B), L(B) = L_\sigma(B)$, and $C = \sigma(A,B), \widehat{C} = \sigma(\widehat{A},\widehat{B})$ in the following texts. Here $K(B)$ is the unique part from $\{B^\circ, B^c\}$ that lies entirely inside or outside $\sigma(A,B)$, thus it makes no contribution to the boundary of $\sigma(A,B)$. And L(B) is the other part (Fig. 7).

Proposition 2.2.2. For $\forall point\ P \in \partial\widehat{A}_i, \partial\widehat{A}_i \subseteq Iso_A$, we have:

$$\begin{aligned} &(1)\quad P \in K(B)\ \ or\ \ P \in L(B) \\ &(2)\quad \begin{aligned} P \in K(B) &\iff \partial\widehat{A}_i \cap \partial\sigma(\widehat{A},\widehat{B}) = \emptyset \\ P \in L(B) &\iff \partial\widehat{A}_i \cap \partial\sigma(\widehat{A},\widehat{B}) = \partial\widehat{A}_i \end{aligned} \end{aligned} \tag{13}$$

Proposition 2.2.3. The definition of ∂C_{Iso} in *Algorithm* 1 satisfies:

$$\partial C_{Iso} = \partial\widehat{C} \cap (Iso_{\widehat{A}} \cup Iso_{\widehat{B}}).$$

Proof. This follows directly from *Proposition 2.2.2.* □

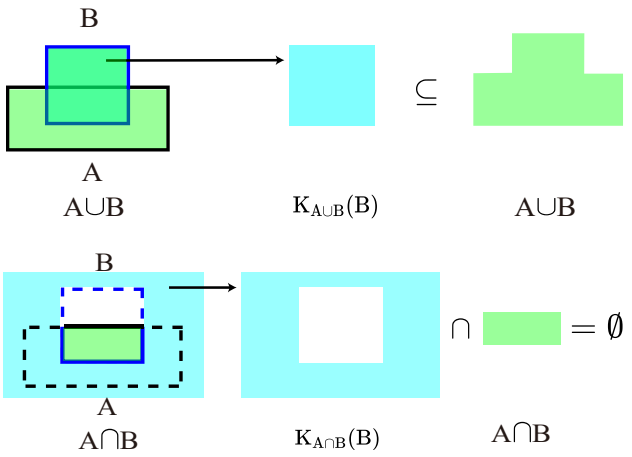


Fig. 7. Illustration of Def 2.2.1.

2.3 Toward Loop-Level Theorem

Previous discussions only consider B-rep gluing at the face level, which may not be sufficient for many real-world cases. For example, in the case shown in Fig. 9, efficient regeneration can be achieved via loop-level gluing, whereas reuse is not available under the face-level method.

In this subsection, we establish notation for describing gluing at the loop level and present our loop-level gluing method. This method provides a preliminary treatment of loop reuse and enables efficient regeneration for a certain class of cases.

Definition 2.3.1. For $\alpha \in F_A$ we define:

$$\alpha^{-}_{\text{gen}} \triangleq \begin{cases} \alpha, & \alpha_{oLoop} \cap \overline{K(B)} \neq \emptyset, \\ \emptyset, & \alpha \cap \overline{K(B)} = \emptyset, \\ \alpha - \langle \alpha_{oLoop} \rangle_{\alpha - \overline{K(B)}}, & \alpha \cap \overline{K(B)} \neq \emptyset,\ \alpha_{oLoop} \cap \overline{K(B)} = \emptyset, \end{cases}$$

$$\alpha^{-}_{\text{diff}} \triangleq \begin{cases} \alpha, & \alpha \subseteq \partial\widehat{A}, \\ \emptyset, & \alpha_{oLoop} \nsubseteq \partial\widehat{A}, \\ \overline{\alpha - \langle \alpha_{oLoop} \rangle_{\alpha \cap \partial\widehat{A}}}, & \alpha \nsubseteq \partial\widehat{A},\ \text{outer loops are invariant in } \widehat{A}. \end{cases}$$

Here, α_{oLoop} denotes the outer loops of α, and $\langle X \rangle_Y$ denotes the connected component in Y that contains X.

Under these notations, we can represent the result of loop-level Boolean operations in *Definition 2.3.2*:

Definition 2.3.2. For $\alpha \in F_A, \beta \in F_B$, we define:

$$\begin{aligned} \alpha^{*}_{gen} &\triangleq \alpha - \alpha^{-}_{gen}, \\ \beta^{*}_{gen} &\triangleq \beta - \beta^{-}_{gen}, \\ gen^{+} &\triangleq F_C - (\cup_{\alpha \in F_A}(\alpha^{*}_{gen})) - (\cup_{\beta \in F_B}(\beta^{*}_{gen})), \\ gen^{-}_{A} &\triangleq \cup_{\alpha \in F_A}(\alpha^{-}_{gen}), \\ gen^{-}_{B} &\triangleq \cup_{\beta \in F_B}(\beta^{-}_{gen}). \end{aligned}$$

Definition 2.3.3. For $\alpha \in F_A, \beta \in F_B$, we define:

Algorithm 1 Boolean for Isolated Components

Inputs: Isolated Components $\partial\widehat{A}_i, \partial\widehat{B}_j$, solid $\widehat{A}, \widehat{B}$
Outputs: Boolean Result from Isolated Components C_{Iso}
Initialize $\partial C_{Iso} \leftarrow \emptyset$
for $\partial\widehat{A}_i$ **do**
    Get point $P_i \in \partial\widehat{A}_i$
    if $PointInRegion\ (P_i, L(\widehat{B}))$ **then**
        add $\partial\widehat{A}_i \rightarrow \partial C_{Iso}$
    end if
end for
for $\partial\widehat{B}_j$ **do**
    Get point $Q_j \in \partial\widehat{B}_j$
    if $PointInRegion\ (Q_j, L(\widehat{A}))$ **then**
        add $\partial\widehat{B}_j \rightarrow \partial C_{Iso}$
    end if
end for
return C_{Iso}

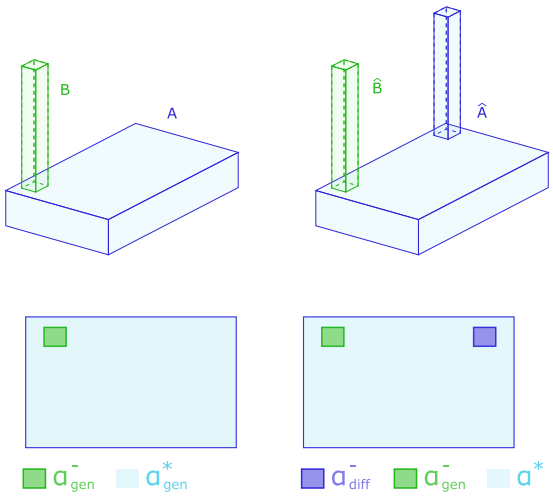


Fig. 8. Illustrations of def 2.3.2.

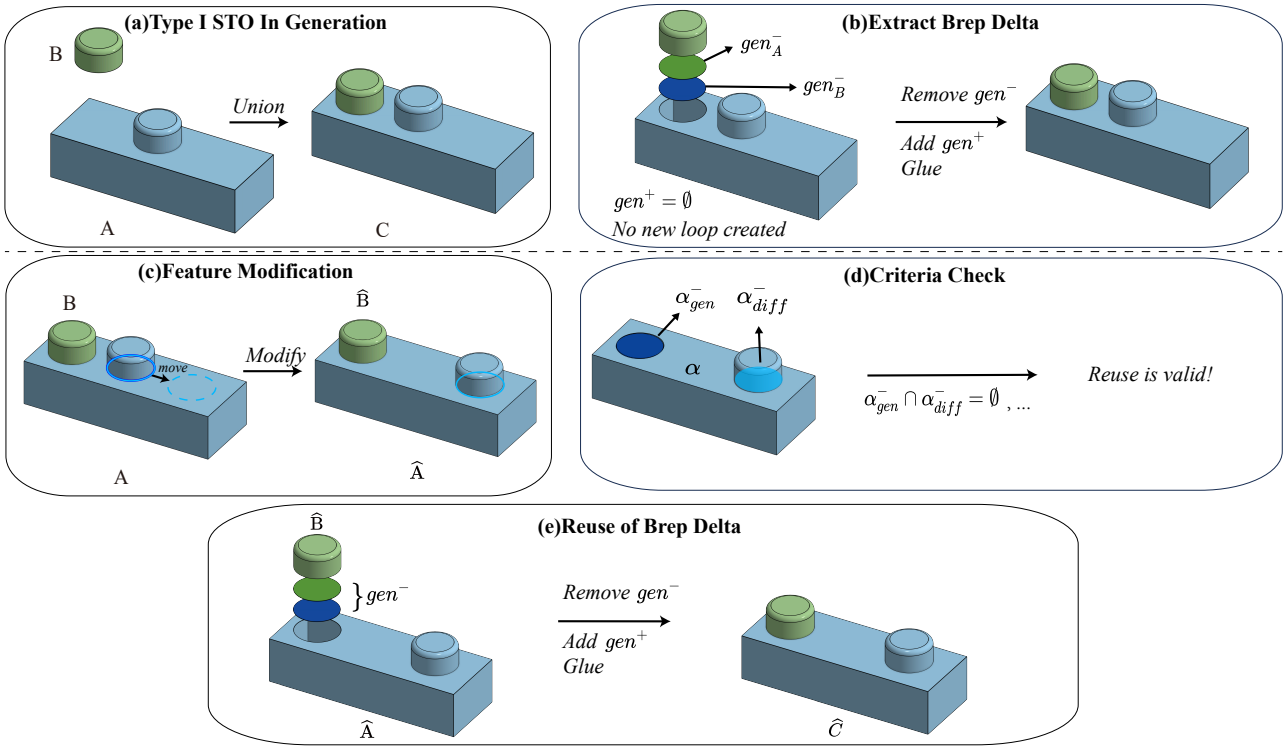


Fig. 9. A LEGO-like example of loop-level B-rep data reuse. (a) A Type-I STO involving Boolean union during generation. (b) The B-rep delta extracted from the Boolean process. (c) Model A is modified by moving the blue protrusion downwards to $\widehat{A}$, which triggers regeneration process. (d) B-reps required in criteria are extracted for check. In this step, the only non-trivial check is shown in the figure. (e) Reuse of saved B-rep data to update the model by gluing.

$$\widehat{\alpha} \triangleq \alpha - \alpha_{diff}^-,$$
$$\widehat{\beta} \triangleq \beta - \beta_{diff}^-,$$
$$\alpha^* \triangleq \widehat{\alpha} - \alpha_{gen}^-,$$
$$\beta^* \triangleq \widehat{\beta} - \beta_{gen}^-,$$
$$newf_A \triangleq \{f \in F_{\widehat{A}} \mid f \neq \widehat{\alpha}, \forall \alpha \in F_A\},$$
$$newf_B \triangleq \{f \in F_{\widehat{B}} \mid f \neq \widehat{\beta}, \forall \beta \in F_B\}.$$

Under these definition, α^*, β^* can be glued at the loop-level. Therefore, it remains to prove validity through the following equation:

$$\partial\widehat{C} = \partial C_{Iso} \cup \Big(\bigcup_{\alpha \in F_A, \widehat{\alpha} \subseteq Int_{\widehat{A}}} \alpha^* \Big) \cup \Big(\bigcup_{\beta \in F_B, \widehat{\beta} \subseteq Int_{\widehat{B}}} \beta^* \Big) \cup \mathrm{gen}^+.$$

By the definitions of α^* and β^*, we have the following proposition.

Proposition 2.3.4.

$$(1)(newf_A \cap Int_{\widehat{A}}) \cup \bigcup_{\alpha \in F_A, \widehat{\alpha} \subseteq Int_{\widehat{A}}} \alpha^* = Int_{\widehat{A}} - \mathrm{gen}_A^-,$$
$$(2)(newf_B \cap Int_{\widehat{B}}) \cup \bigcup_{\beta \in F_B, \widehat{\beta} \subseteq Int_{\widehat{B}}} \beta^* = Int_{\widehat{B}} - \mathrm{gen}_B^-.$$

Thus, our gluing result is identical to

$$\partial C_{Iso} \cup \left(Int_{\widehat{A}} - \mathrm{gen}_A^-\right) \cup \left(Int_{\widehat{B}} - \mathrm{gen}_B^-\right) \cup \mathrm{gen}^+.$$

which is analogous to the face-level case.

3 Theory in Generic Cases

3.1 Criteria

Criterion 1
$$\forall \alpha \in F_A, \quad \alpha_{gen}^- \cap \alpha_{diff}^- = \emptyset,$$
$$\forall \beta \in F_B, \quad \beta_{gen}^- \cap \beta_{diff}^- = \emptyset.$$

Criterion 2
$$\forall f \in \mathrm{newf}_A, \quad f \cap \partial\widehat{B} = \emptyset,$$
$$\forall f \in \mathrm{newf}_B, \quad f \cap \partial\widehat{A} = \emptyset.$$

3.2 Lemmas

Proposition 3.2.1.

$$U(P, \epsilon_P) \cap \widehat{A} = U(P, \epsilon_P) \cap A,$$
$$P \in \partial A \cap \partial B \Rightarrow \exists \epsilon_P > 0, \ s.t. \ U(P, \epsilon_P) \cap \widehat{B} = U(P, \epsilon_P) \cap B,$$
$$U(P, \epsilon_P) \cap \widehat{C} = U(P, \epsilon_P) \cap C.$$

Proof. For any face $\alpha \in F_A$, we consider the following two cases.

(1) If $P \in \alpha$:

Since $P \in \partial B$, by the definition of α_{gen}^-, we have $P \in \alpha_{\mathrm{gen}}^-$. By *Criterion 1*, we have $d(P, \alpha_{\mathrm{diff}}^-) > 0$. Moreover, α_{diff}^- is closed by definition.

Thus, there exists $\epsilon_P^\alpha > 0$ such that

$$U(P, \epsilon_P^\alpha) \cap \alpha^-_{\text{diff}} = \emptyset.$$

As a consequence,

$$U(P, \epsilon_P^\alpha) \cap \alpha \subseteq \widehat{\alpha}.$$

(2) If $P \notin \alpha$:

We have $d(P, \alpha) > 0$. Let $\epsilon_P^\alpha \triangleq \frac{1}{2} d(P, \alpha) > 0$. Then

$$U(P, \epsilon_P^\alpha) \cap \alpha = \emptyset \subseteq \widehat{\alpha}.$$

Combining (1) and (2), define $\epsilon_P^A \triangleq min_{\alpha \in F_A}\{\epsilon_P^\alpha\}$. Then, for every $\alpha \in F_A$, we have

$$\alpha \cap U(P, \epsilon_P^A) \subseteq \widehat{\alpha}.$$

By symmetry, the similar conclusion holds for B.
Now consider a face $f \in F_{\widehat{A}}$.

(i) If $\exists \alpha \in F_A \; s.t. \; f = \widehat{\alpha}$:

Since $\widehat{\alpha} \subseteq \alpha$, there exists $\epsilon_P^f > 0$ such that $U(P, \epsilon_P^f) \cap \widehat{\alpha} \subseteq \alpha$.

(ii) If $\forall \alpha \in F_A, f \neq \widehat{\alpha}$:

Then $f \in newf_A$. By *Criterion 2*,

$$f \in newf_A \Longrightarrow f \cap \partial\widehat{B} = \emptyset \Longrightarrow P \notin f.$$

Hence, there exists $\epsilon_P^f = \frac{1}{2} d(P, f) > 0$ such that

$$U(P, \epsilon_P^f) \cap f = \emptyset.$$

Combining (i) and (ii), define $\epsilon_A \triangleq min\{\epsilon_P^\alpha, \epsilon_P^f\}$. Then F_A and $F_{\widehat{A}}$ coincide completely within $U(P, \epsilon_A)$, which implies

$$U(P, \epsilon_A) \cap \widehat{A} = \; U(P, \epsilon_A) \cap A. \tag{14}$$

Similarly, there exists $\epsilon_B > 0$ such that

$$U(P, \epsilon_B) \cap \widehat{B} = U(P, \epsilon_B) \cap B.$$

Define $\epsilon_P \triangleq \min(\epsilon_A, \epsilon_B)$. Then the conclusion follows from

$$C = A \cup B, \qquad \widehat{C} = \widehat{A} \cup \widehat{B}.$$

□

Lemma 3.2.1.

$$\partial A \cap \partial B = \partial\widehat{A} \cap \partial\widehat{B}.$$

Proof. By *Proposition 3.2.1*, we directly have

$$\partial A \cap \partial B \subseteq \partial\widehat{A} \cap \partial\widehat{B}. \tag{15}$$

It remains to prove $\partial\widehat{A} \cap \partial\widehat{B} \subseteq \partial A \cap \partial B$. Suppose that

$$P \in \partial\widehat{A} \cap \partial\widehat{B}, \qquad P \notin \partial A \cap \partial B. \tag{16}$$

By symmetry, we may assume $P \notin \partial A$. Then

$$\begin{aligned} &P \in \partial\widehat{A} - \partial A \subseteq newf_A, \\ &P \in \partial\widehat{A} \cap \partial\widehat{B} \subseteq \partial\widehat{B}, \end{aligned} \tag{17}$$

which contradicts *Criterion 2*.
Therefore,

$$P \in \partial\widehat{A} \cap \partial\widehat{B} \Longrightarrow P \in \partial A \cap \partial B.$$

This proves the result.

□

Lemma 3.2.2.

$$\partial\widehat{A} \cap \partial\widehat{B} \cap \partial\widehat{C} = \partial A \cap \partial B \cap \partial C.$$

Proof. Directly follows from *Proposition 3.2.1* and *Lemma 3.2.1*.

□

Lemma 3.2.3.

$$Int_{\widehat{A}} \cap K(\widehat{B}) = Int_A \cap K(B).$$

Proof. For $\forall P \in Int_{\widehat{A}} \cap K(\widehat{B})$, we know $\exists X_0 \in \partial\widehat{A} \cap \partial\widehat{B}$ such that

$$\gamma : [0, 1] \to \partial\widehat{A}, \quad \begin{aligned} &\gamma(0) = P, \\ &\gamma(1) = X_0. \end{aligned}$$

Moreover, we may assume that $\gamma([0, 1)) \cap \partial\widehat{B} = \emptyset$. Otherwise we can let X^* be the first intersection point between γ and $\partial\widehat{B}$, and let $\widehat{\gamma}$ be the path from P to X^*. Then we have

$$\gamma(0) \in K(\widehat{B}) \;\Rightarrow\; \gamma([0, 1)) \subseteq K(\widehat{B}).$$

From *Proposition 3.2.1*, we know that there exist $\epsilon > 0$ and an open neighborhood U such that

$$\gamma((1-\epsilon, 1)) \subseteq U, \quad \begin{aligned} A \cap U &= \widehat{A} \cap U, \\ B \cap U &= \widehat{B} \cap U, \\ C \cap U &= \widehat{C} \cap U. \end{aligned}$$

Therefore,

$$\begin{aligned} \gamma((1-\epsilon, 1)) \subseteq \partial\widehat{A} \Rightarrow \gamma((1-\epsilon, 1)) \subseteq \partial A, \\ \gamma((1-\epsilon, 1)) \subseteq K(\widehat{B}) \Rightarrow \gamma((1-\epsilon, 1)) \subseteq K(B). \end{aligned}$$

Define

$$\begin{aligned} S_1 &\triangleq \{\, t \in [0, 1) \mid \gamma(t) \notin \partial A \,\}, \\ S_2 &\triangleq \{\, t \in [0, 1) \mid \gamma(t) \notin K(B) \,\}. \end{aligned}$$

If $S_1 \cup S_2 \neq \varnothing$, let $t_0 = \sup(S_1 \cup S_2) \leq 1 - \epsilon$:

(1) $t_0 \in S_1 (fig.\ 3)$:

By the definition of S_1, we have

$$\gamma(t_0) \in \partial\widehat{A} \cap \partial A \cap \overline{\partial\widehat{A} - \partial A}.$$

Then, by *Proposition 3.2.2*, we obtain

$$\gamma(t_0) \in \overline{\partial A - \partial\widehat{A}}.$$

Consequently, there exists $\alpha \in F_A$ such that $\gamma(t_0) \in \alpha$ and $\alpha \notin F_{\widehat{A}}$. Moreover,

$$\sup S_2 \leq t_0 \Longrightarrow \gamma((t_0, 1)) \subseteq K(B) \Longrightarrow \gamma(t_0) \in \overline{K(B)}.$$

Now,

$$\gamma(t_0) \in \alpha \cap \overline{K(B)} \Longrightarrow \alpha \in \text{gen}_A^- \overset{\text{Criterion 1}}{\Longrightarrow} \alpha \in F_{\widehat{A}},$$

which yields a contradiction.

(2) $t_0 \in S_2 (fig.\ 4)$:

In this case, by the definition of S_2, we have $\gamma(t_0) \in \partial B$. Moreover,

$$\sup S_1 \le t_0 \Longrightarrow \gamma((t_0, 1)) \subseteq \partial A \Longrightarrow \gamma(t_0) \in \partial A.$$

Then, by *Lemma 3.2.1*, we have

$$\gamma(t_0) \in \partial A \cap \partial B = \partial\widehat{A} \cap \partial\widehat{B},$$

which contradicts our previous assumption that $\gamma([0,1)) \cap \partial\widehat{B} = \emptyset$.

From the above two cases, we conclude that $S_1 \cup S_2 = \varnothing$. Thus, $P \in \partial A \cap K(B)$. Moreover, there exists a path γ from P to $X_0 \in \partial A \cap \partial B$ such that $\gamma([0,1]) \subseteq \partial A$. Hence, $P \in Int_A$, and therefore

$$Int_{\widehat{A}} \cap K(\widehat{B}) \subseteq Int_A \cap K(B).$$

For $\forall Q \in Int_A \cap K(B)$, let $Q \in \alpha_0 \in F_A$. Then, by *Criterion 1*, we have $\alpha_0 \in \text{gen}_{\widehat{A}}^- \subseteq F_{\widehat{A}}$, and hence $Q \in \partial\widehat{A}$.

There exists $X_0 \in \partial A \cap \partial B$ such that

$$\gamma : [0,1] \to \partial A, \quad \begin{matrix} \gamma(0) = Q, \\ \gamma(1) = X_0. \end{matrix}$$

Moreover, we may assume that $\gamma([0,1)) \cap \partial B = \emptyset$. Otherwise, we can let X^* be the first intersection point between γ and ∂B, and let $\widehat{\gamma}$ be the path from Q to X^*. Then we have

$$\gamma(0) \in K(B) \Longrightarrow \gamma([0,1)) \subseteq K(B).$$

From *Proposition 3.2.1*, we know that there exist $\epsilon > 0$ and an open neighborhood U such that

$$\gamma((1-\epsilon, 1)) \subseteq U, \quad \begin{matrix} A \cap U = \widehat{A} \cap U, \\ B \cap U = \widehat{B} \cap U, \\ C \cap U = \widehat{C} \cap U. \end{matrix}$$

Therefore,

$$\gamma((1-\epsilon,1)) \subseteq \partial A \Longrightarrow \gamma((1-\epsilon,1)) \subseteq \partial\widehat{A},$$
$$\gamma((1-\epsilon,1)) \subseteq K(B) \Longrightarrow \gamma((1-\epsilon,1)) \subseteq K(\widehat{B}).$$

Define

$$S_1 \triangleq \{\, t \in [0,1) \mid \gamma(t) \notin \partial\widehat{A} \,\},$$
$$S_2 \triangleq \{\, t \in [0,1) \mid \gamma(t) \notin K(\widehat{B}) \,\}.$$

If $S_1 \cup S_2 \neq \varnothing$, let $t_0 = \sup(S_1 \cup S_2) \le 1 - \epsilon$:

(1) $t_0 \in S_1$:

Then there exists $\alpha \in F_A$ such that $\gamma(t_0) \in \alpha$ and $\alpha \notin F_{\widehat{A}}$. Hence,

$$\gamma(t_0) \in \alpha \cap \overline{K(B)} \Longrightarrow \alpha \in \text{gen}_{\widehat{A}}^- \overset{\text{Criterion 1}}{\Longrightarrow} \alpha \in F_{\widehat{A}},$$

which yields a contradiction.

(2) $t_0 \in S_2$:

In this case, by the definition of S_2, we have $\gamma(t_0) \in \partial\widehat{B}$. Moreover,

$$\sup S_1 \le t_0 \Longrightarrow \gamma((t_0,1)) \subseteq \partial\widehat{A} \Longrightarrow \gamma(t_0) \in \partial\widehat{A}.$$

Therefore,

$$\gamma(t_0) \in \partial\widehat{A} \cap \partial\widehat{B} = \partial A \cap \partial B$$

by *Lemma 3.2.1*, which contradicts our previous assumption that $\gamma([0,1)) \cap \partial B = \emptyset$.

From the above two cases, we conclude that $S_1 \cup S_2 = \varnothing$. Thus, $Q \in \partial\widehat{A} \cap K(\widehat{B})$.

Moreover, γ is a path from Q to $X_0 \in \partial A \cap \partial B$, and $\gamma([0,1]) \subseteq \partial\widehat{A}$. Hence, $Q \in Int_{\widehat{A}}$, and therefore

$$Int_A \cap K(B) \subseteq Int_{\widehat{A}} \cap K(\widehat{B}).$$

□

Proposition 3.2.4.

$$gen^+ \cap \partial A \subseteq gen_A^-. \tag{18}$$

Proof. We prove this proposition by contradiction. If $gen^+ \cap \partial A \nsubseteq gen_A^-$, then $\exists P \in gen^+ \cap \partial A$ such that $P \notin gen_A^-$.

Then assume that the point P lies on faces α_i $(1 \le i \le m)$ in F_A and faces β_j $(1 \le j \le n)$ in F_B. Since $P \notin gen_A^-$, we know that $\alpha_i \cap B = \emptyset$, and hence $\alpha_i \in F_C$, $\alpha_i \notin gen^+$. Since $P \in gen^+$, there exists a face γ such that $P \in \gamma$ and $\gamma \in gen^+$. Then, from the definitions of α_i and β_j, we know that there exists an open neighborhood U of P such that

$$U \cap \partial A = U \cap (\bigcup_{i=1}^{m} \alpha_i), \qquad U \cap \partial B = U \cap (\bigcup_{j=1}^{n} \beta_j).$$

Moreover,

$$U \cap \gamma \subseteq U \cap \partial C \subseteq U \cap (\partial A \cup \partial B) \subseteq U \cap ((\bigcup_{i=1}^{m} \alpha_i) \cup (\bigcup_{j=1}^{n} \beta_j)).$$

Let m_2 denote the 2-dimensional measure induced on ∂C. Then

$$\begin{aligned} 0 &< m_2(U \cap \gamma) \\ &\le \sum_{i=1}^{m} m_2(U \cap \gamma \cap \alpha_i) + \sum_{j=1}^{n} m_2(U \cap \gamma \cap \beta_j) \\ &\le \sum_{i=1}^{m} m_2(\gamma \cap \alpha_i) + \sum_{j=1}^{n} m_2(\gamma \cap \beta_j). \end{aligned}$$

Since $\alpha_i \notin gen^+$ and $\gamma \in gen^+$, we have $\alpha_i \neq \gamma$ for all i. Recalling that $\alpha_i, \gamma \in F_C$, α_i and γ are distinct faces in the valid B-rep model C. Thus, their intersection is at most 1-dimensional, and $m_2(\alpha_i \cap \gamma) = 0$. Consequently,

$$0 < \sum_{i=1}^{m} m_2(\gamma \cap \alpha_i) + \sum_{j=1}^{n} m_2(\gamma \cap \beta_j) = \sum_{j=1}^{n} m_2(\gamma \cap \beta_j).$$

Therefore, we must have $n > 0$, which implies

$$P \in \partial B, \qquad P \in \partial A \cap \partial B \Rightarrow P \in gen_A^-,$$

which leads to a contradiction. □

3.3 Main Theorem in Generic Cases

We first state the main result:

Theorem 3.3.1.

$$\partial\widehat{C} = \partial C_{Iso} \cup \left(Int_{\widehat{A}} - \text{gen}_{\widehat{A}}^-\right) \cup \left(Int_{\widehat{B}} - \text{gen}_{\widehat{B}}^-\right) \cup \text{gen}^+.$$

Proof. Because $Iso_{\widehat{A}}, Iso_{\widehat{B}}, Int_{\widehat{A}} \cup Int_{\widehat{B}}$ does not intersect with each other, we can decompose $\partial\widehat{C}$ into three parts:

$$\partial\widehat{C} = (\partial\widehat{C} \cap Iso_{\widehat{A}}) \cup (\partial\widehat{C} \cap Iso_{\widehat{B}}) \cup (\partial\widehat{C} \cap (Int_{\widehat{A}} \cup Int_{\widehat{B}})).$$

Moreover, by *Proposition 2.2.3* we have

$$\partial C_{Iso} = \partial\widehat{C} \cap (Iso_{\widehat{A}} \cup Iso_{\widehat{B}}).$$

It therefore suffices to prove

$$\partial\widehat{C} \cap (Int_{\widehat{A}} \cup Int_{\widehat{B}}) = \left(Int_{\widehat{A}} - \mathrm{gen}_{\widehat{A}}^{-}\right) \cup \left(Int_{\widehat{B}} - \mathrm{gen}_{\widehat{B}}^{-}\right) \cup \mathrm{gen}^{+}.$$

By symmetry, it is enough to establish the following two inclusions, which we record as Propositions 3.3.2 and 3.3.3:

$$(1) \quad \partial\widehat{C} \cap Int_{\widehat{A}} \subseteq (Int_{\widehat{A}} - \mathrm{gen}_{\widehat{A}}^{-}) \cup \mathrm{gen}^{+}.$$
$$(2) \quad (Int_{\widehat{A}} - \mathrm{gen}_{\widehat{A}}^{-}) \cup (\mathrm{gen}^{+} \cap \partial A) \subseteq \partial\widehat{C} \cap Int_{\widehat{A}}.$$

□

Proposition 3.3.2.

$$\partial\widehat{C} \cap Int_{\widehat{A}} \subseteq (Int_{\widehat{A}} - \mathrm{gen}_{\widehat{A}}^{-}) \cup \mathrm{gen}^{+}.$$

Proof. We first show that

$$\partial\widehat{C} \cap Int_{\widehat{A}} \subseteq \partial C.$$

Decomposing with respect to $K(B)$, ∂B, and $L(B)$. Then

$$\begin{aligned}(\partial\widehat{C} \cap Int_{\widehat{A}} \cap \partial A) \cap K(B) &\subseteq \partial\widehat{C} \cap \partial\widehat{A} \cap (\partial A \cap K(B)) \\ &= \partial\widehat{C} \cap \partial\widehat{A} \cap (\partial\widehat{A} \cap K(\widehat{B})) = \emptyset. \qquad (\textit{Lemma 3})\end{aligned}$$

$$\begin{aligned}(\partial\widehat{C} \cap Int_{\widehat{A}} \cap \partial A) \cap \partial B &\subseteq \partial\widehat{C} \cap \partial\widehat{A} \cap \partial A \cap \partial B \\ &= \partial\widehat{C} \cap \partial\widehat{A} \cap \partial\widehat{B} \qquad (\textit{Lemma 1}) \\ &= \partial C \cap \partial A \cap \partial B \subseteq \partial C. \qquad (\textit{Lemma 2})\end{aligned}$$

$$(\partial\widehat{C} \cap Int_{\widehat{A}} \cap \partial A) \cap L(B) \subseteq \partial A \cap L(B) \subseteq \partial C.$$

Therefore,

$$\partial\widehat{C} \cap Int_{\widehat{A}} \cap \partial A \subseteq \partial C \;\Rightarrow\; \partial\widehat{C} \cap Int_{\widehat{A}} \cap (\partial A - \partial C) = \emptyset.$$

Since $\partial\widehat{C} \cap Int_{\widehat{A}} \subseteq Int_{\widehat{A}}$, this implies

$$\partial\widehat{C} \cap Int_{\widehat{A}} \subseteq Int_{\widehat{A}} - (\partial A - \partial C).$$

By the definition of the B-rep delta,

$$\mathrm{gen}_{\widehat{A}}^{-} - \mathrm{gen}^{+} \subseteq \partial A - \partial C.$$

Hence,

$$\begin{aligned}\partial\widehat{C} \cap Int_{\widehat{A}} &\subseteq Int_{\widehat{A}} - (\partial A - \partial C) \\ &\subseteq Int_{\widehat{A}} - \left(\mathrm{gen}_{\widehat{A}}^{-} - \mathrm{gen}^{+}\right) \\ &\subseteq (Int_{\widehat{A}} - \mathrm{gen}_{\widehat{A}}^{-}) \cup \mathrm{gen}^{+}.\end{aligned}$$

□

Proposition 3.3.3.

$$(Int_{\widehat{A}} - \mathrm{gen}_{\widehat{A}}^{-}) \cup (\mathrm{gen}^{+} \cap \partial A) \subseteq \partial\widehat{C} \cap Int_{\widehat{A}}.$$

Proof. It suffices to prove that the two parts on the left belong to the right.

First, we prove that $Int_{\widehat{A}} - gen_{\widehat{A}}^{-} \subseteq \partial\widehat{C} \cap Int_{\widehat{A}}$. By the definition of $gen_{\widehat{A}}^{-}$, we have

$$Int_A \cap \overline{K(B)} \subseteq gen_{\widehat{A}}^{-},$$

Then, by *Lemma 3.2.1* and *Lemma 3.2.3*,

$$\begin{aligned}Int_{\widehat{A}} - gen_{\widehat{A}}^{-} &\subseteq Int_{\widehat{A}} - (Int_A \cap \overline{K(B)}) \\ &= Int_{\widehat{A}} - (Int_{\widehat{A}} \cap \overline{K(\widehat{B})}) \\ &= Int_{\widehat{A}} \cap L(\widehat{B}) \subseteq \partial\widehat{C} \cap Int_{\widehat{A}}.\end{aligned}$$

Second, we prove that $gen^{+} \cap \partial A \subseteq \partial\widehat{C} \cap Int_{\widehat{A}}$. By *Proposition 3.2.4* and *Criterion 1*,

$$\begin{aligned}gen^{+} \cap \partial A &\subseteq gen_{\widehat{A}}^{-} \\ &\subseteq Int_{\widehat{A}}.\end{aligned}$$

This implies that

$$gen^{+} \cap \partial A = gen^{+} \cap \partial A \cap Int_{\widehat{A}}.$$

Then, by *Lemma 3.2.3*,

$$\begin{aligned}gen^{+} \cap \partial A \cap Int_{\widehat{A}} \cap K(\widehat{B}) &= gen^{+} \cap \partial A \cap Int_A \cap K(B) \\ &\subseteq \partial C \cap \partial A \cap K(B) = \emptyset.\end{aligned}$$

By *Lemma 3.2.1* and *Lemma 3.2.2*,

$$\begin{aligned}gen^{+} \cap \partial A \cap Int_{\widehat{A}} \cap \partial\widehat{B} &\subseteq \partial C \cap \partial A \cap \partial\widehat{A} \cap \partial\widehat{B} \\ &= \partial C \cap \partial A \cap \partial A \cap \partial B \\ &= \partial\widehat{C} \cap \partial\widehat{A} \cap \partial\widehat{B} \\ &\subseteq \partial\widehat{C}.\end{aligned}$$

Finally, by *Lemma 3.2.2* again,

$$gen^{+} \cap \partial A \cap Int_{\widehat{A}} \cap L(\widehat{B}) \subseteq Int_{\widehat{A}} \cap L(\widehat{B}) \subseteq \partial\widehat{C}.$$

Combining the above results, we obtain $gen^{+} \cap \partial A \subseteq \partial\widehat{C} \cap Int_{\widehat{A}}$, which completes the proof. □

4 Other Operations

CAD systems support a wide variety of operations beyond Boolean ones, making it impractical to discuss every operation in detail here. Instead, we demonstrate the applicability of the proposed method using a simple edge fillet example.

We denote this operation by

$$Fillet(e, r),$$

which generates the solid C from A by applying a fillet of radius r to edge e.

$$\begin{aligned}A = M_{k-1}, \quad C = M_k = S_k(M_{k-1}, e, r) = Fillet(A, e, r).\\ \widehat{A} = \widehat{M}_{k-1}, \quad \widehat{C} = \widehat{M}_k = S_k(\widehat{M}_{k-1}, \widehat{e}, \widehat{r}) = Fillet(\widehat{A}, \widehat{e}, \widehat{r}).\end{aligned}$$

When $\widehat{e} \neq e$ or $\widehat{r} \neq r$, the reuse of B-rep delta is impossible and we fall back to recomputation of the fillet. In the following discussion, we only consider the cases where

$$\widehat{e} = e \quad \text{and} \quad \widehat{r} = r.$$

4.1 Definitions

Definition 4.1.1.

$$\begin{aligned}gen^{-} &= \{f \in F_A \mid f \text{ is incident to } e\} \\ gen^{+} &= F_C \setminus (F_A \setminus gen^{-})\end{aligned} \tag{19}$$

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	20
TR(%)	80.87	52.75	50.79	80.33	71.26	70.86	76.96	76.54	72.19	35.80	52.54	49.40	30.14	92.65	56.40	52.21	48.77	44.36	100.03
SR(%)	22.22	63.89	65.71	23.53	39.39	40.63	29.03	30.00	41.38	82.14	62.96	69.23	92.00	16.67	56.52	63.64	71.43	80.00	0.00

Table 1. TR and SR of each modification.

Definition 4.1.2.

$$newf = F_{\widehat{A}} \setminus F_A \tag{20}$$

4.2 Criteria

4.2.1 Criterion 1.

$$gen^- \subseteq F_{\widehat{A}} \tag{21}$$

4.2.2 Criterion 2.

$$newf \cap gen^+ = \emptyset \tag{22}$$

Since the geometric effect of the fillet operation is completely determined by the supporting faces in gen^-, Criterion 1 guarantees that the induced delta face set remains invariant during regeneration, thereby validating the reuse of the stored delta. Criterion 2 further guarantees that the reused delta does not conflict with newly generated faces, ensuring that the resulting glued model forms a valid B-rep. Fig. 14 illustrates the extraction of the B-rep delta and the corresponding criteria checks for the fillet operation. In fact, if a Type-II STO, such as a fillet operation, is formulated as a Boolean process in which an intermediate solid is generated and then combined with the input solid, the treatment described in the main text applies without modification.

5 Supplementary Experiments on Larger Models

In the main text, we verified through our dataset that our algorithm has significant efficiency for features with locality. But those models are too simple to verify the effect of our algorithm on real models. In this section, we conduct experiments on two larger examples to estimate the performance of our algorithm on practical models.

Parameter Modification. we select each feature (except "Mirrored" feature and the last feature, since the "Mirrored" operation is not frequently modified and the last feature has no downstream features) from the beginning to the end and increase its first length parameter by 0.1mm for Type-I operations and 0.01mm for Type-II operations. Information concerning the first feature will not be counted because it does not have previous features and will not trigger our algorithm. Methods for validity evaluation and symbols for efficiency evaluation will be the same as in Section 6 of the main text.

5.1 Model I

The first test model is an industrial part, consisting of 21 features and 37 STOs (In our experiment, "Mirrored" is operated on every selected feature one by one and each mirrored feature is regarded as an STO). The modeling sequence can be seen in Table. 2 and its B-rep in Fig. 10. The VAR time and FBR time consumed by each modification is shown in Fig. 15 and TR is shown in Table. 1. The composition of GR and BR is provided in Fig. 16 and SR of each modification is in Table. 1.

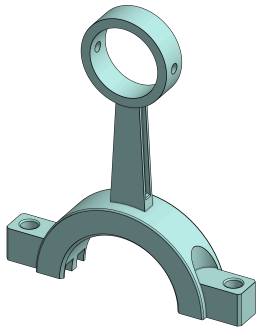


Fig. 10. Model I. The model is adapted from the test model presented by [Aranburu et al. 2022] and was independently reconstructed and modified for our experiments.

index	feature	index	feature	index	feature
1	Pad	8	Fillet001	15	Pocket004
2	Pocket	9	Pocket002	16	Fillet003
3	Pocket001	10	Pocket003	17	Fillet004
4	Pad001	11	Chamfer002	18	Fillet005
5	Chamfer	12	Chamfer003	19	Mirrored
6	Chamfer001	13	Fillet002	20	Pad003
7	Fillet	14	Pad002	21	Pocket005

Table 2. Feature Sequence of Model I.

5.2 Model II

The second test model is a more complex little robot model, consisting of 36 features and 53 STOs. The modeling sequence can be seen in Table. 3 and its B-rep in Fig. 11. The VAR time and FBR time consumed by each modification is shown in Fig. 17 and TR is shown in Table. 5. The composition of GR and BR can be seen in Fig. 18 and SR in Table. 5.

6 A Brief Note on Persistent Naming

During the modeling process, each topological entity—face, edge, and vertex—is assigned a persistent name. On the one hand, these names serve to distinguish different entities; on the other hand, they enable the identification of corresponding or similar topological entities through persistent-name matching.

In FreeCAD, for example, version 0.22.0 dev [FreeCAD contributors 2024] integrates its own naming system, which is able to assign unique names to all topological entities of a solid. Each name is generated according to the way the corresponding topology is created. For instance, every side face produced by an extrusion is swept from

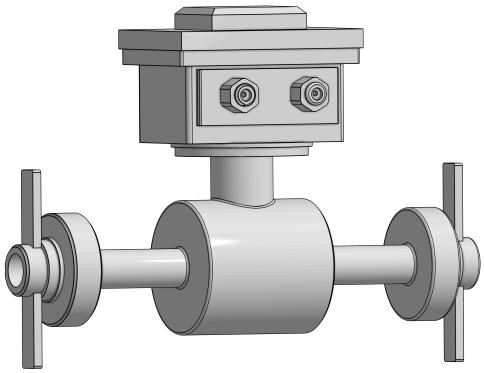


Fig. 11. Model II. The model is adapted from a publicly available GrabCAD model [Hariharavelan 2025] and modified for our experiments.

index	feature	index	feature	index	feature
1	Pad	15	Mirrored	29	Chamfer
2	Pad001	16	Pad007	30	Pad015
3	Pad002	17	Pad008	31	Chamfer001
4	Pad003	18	Pocket001	32	Pad016
5	Pad004	19	Fillet006	33	Fillet010
6	Pad005	20	Pad009	34	Pocket002
7	Pad006	21	Pad010	35	Pocket003
8	Pocket	22	Pad011	36	Mirrored001
9	Fillet	23	Pad012		
10	Fillet001	24	Fillet007		
11	Fillet002	25	Fillet008		
12	Fillet003	26	Fillet009		
13	Fillet004	27	Pad013		
14	Fillet005	28	Pad014		

Table 3. Feature Sequence of Model II.

an edge of the sketch. Therefore, as long as the sketch edges themselves carry unique names, such as $g_1, g_2, \ldots$, the faces generated by the extrusion can also be named uniquely. Fig. 12 briefly illustrates the names assigned to topological entities in FreeCAD.

In practice, we implement Criterion 1 using $\text{gen}_M^- \cap \text{delf}_M = \emptyset$ instead of $\text{gen}_M^- \subseteq F_{\widehat{M}}$, because gen_M^- and delf_M are face sets from the same model M, so their intersection can be checked directly through persistent-naming-set intersection. In contrast, $F_{\widehat{M}}$ belongs to a different model state, so $\text{gen}_M^- \subseteq F_{\widehat{M}}$ does not admit the same convenient implementation.

7 Additional Model Visualizations and Statistics

Fig. 19a presents representative models that exhibit strong locality. The experiments in Section 6 of the main text were conducted on this category of models. Fig.19b shows examples of models we excluded from the DeepCAD dataset [Wu et al. 2021].

In our experiments, we performed five modifications on each feature across all 424 models, resulting in a total of 16,620 modifications (edit-level samples) and 43,019 steps (step-level samples). Detailed statistics are provided below.

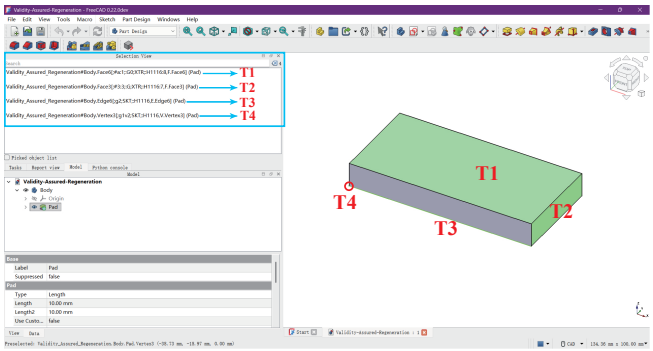


Fig. 12. Name display in the FreeCAD GUI. A basic requirement of any persistent naming algorithm is that all topological entities in a single model receive distinct names; otherwise, name-based references to topology become ambiguous.

Step-Level. For each step in the regeneration, if GR was executed successfully, we recorded the values of t_{c1}, t_{c2}, t_{glue}, t_{VAR}, t_{BR}, and TR. The results are summarized as follows.

	$t_{c1}(\mu s)$	$t_{c2}(\mu s)$	$t_{glue}(\mu s)$	$t_{VAR}(\mu s)$	$t_{BR}(\mu s)$	$TR(\%)$
avg	18.91	23.59	2282.72	2325.22	8937.90	26.37
median	17.00	17.00	2050.00	2096.00	8120.00	25.94

Table 4. Step-level Statistics

Edit-Level. For each edit we recorded global t_{VAR}, t_{FBR}, TR and SR. The average of TR_{global} is 61.70% and the median is 54.32%. Fig. 13 shows the distribution of SR across different total steps.

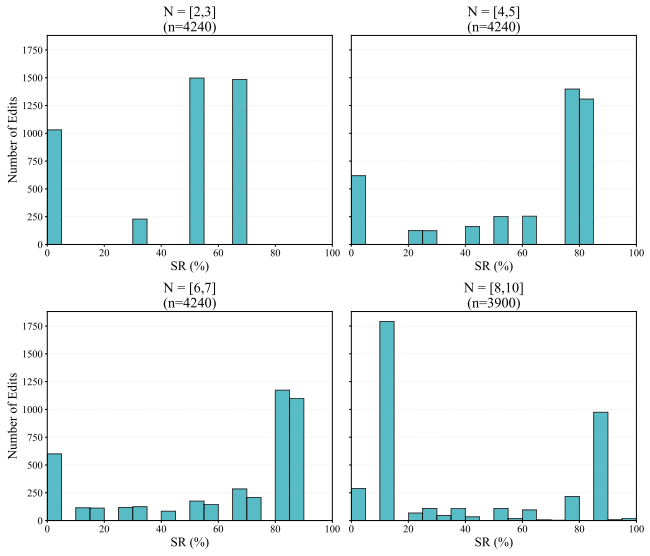


Fig. 13. SR distribution by total step groups.

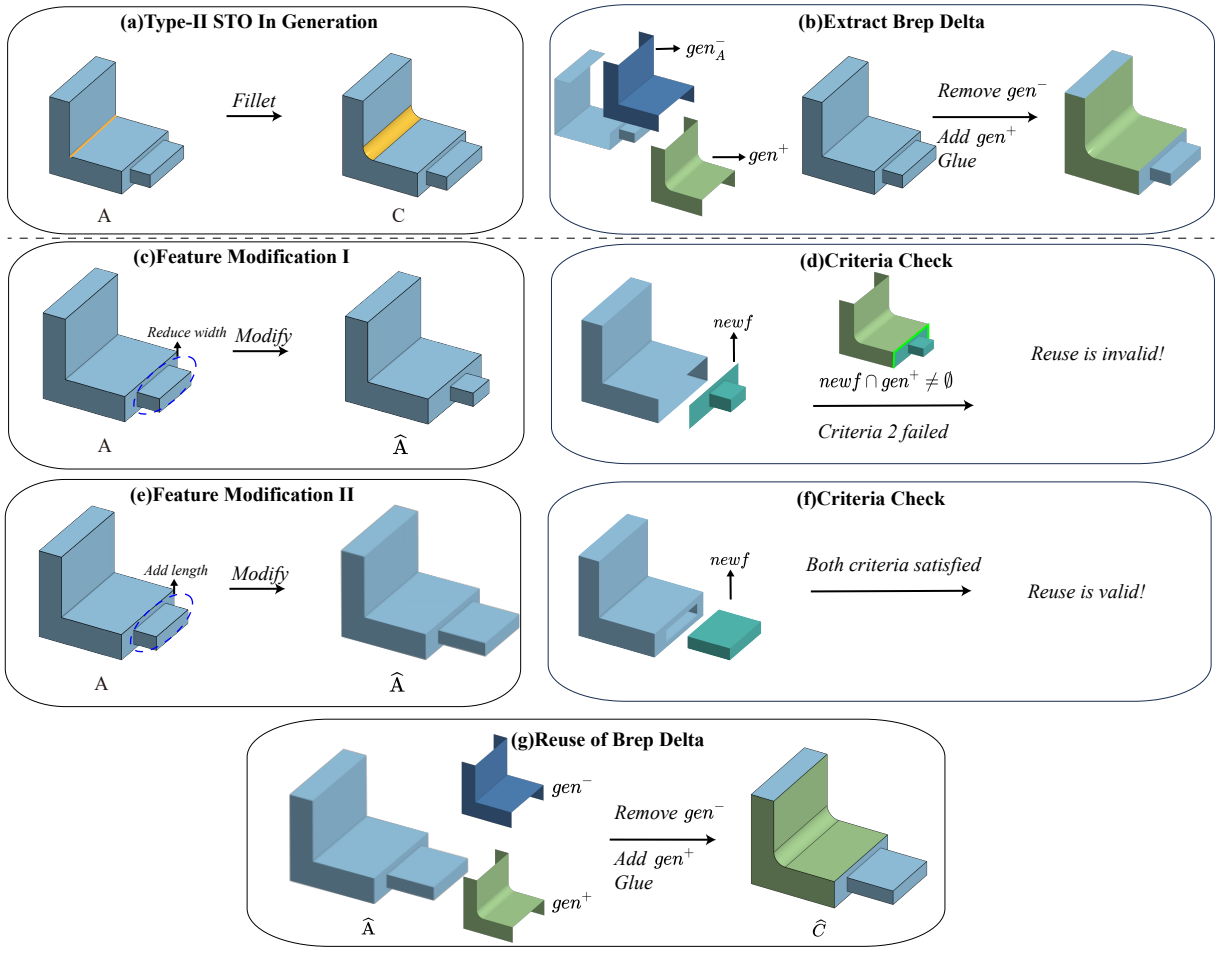


Fig. 14. An example of B-rep check and criteria check for fillet(Type-II STO). (a) During Generation, fillet feature is applied on the yellow edge of A. (b) B-rep delta extracted during generation process. (c) The width of marked protrusion is reduced, triggering regeneration process. (d) $newf$ is calculated for criteria check, and $Criteria\ 2$ failed, meaning reuse is invalid. It is worth noting that a direct reuse also produce the correct $\widehat{C}$, and this issue can be addressed in the loop-level reuse discussed above, where the fillet can pass $Criteria\ 2$. (e) In another regeneration process for same model A, the length of the marked protrusion on A is added. (f) Both criteria in this case are satisfied, and the fillet can be regenerated by GR. (g) The gluing process for the modification in (e).

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	16	17	18
TR(%)	51.46	52.47	50.34	48.99	38.88	46.78	44.90	39.46	44.98	37.18	34.72	31.54	27.45	27.79	62.26	63.64	28.42
SR(%)	42.31	42.31	54.90	56.00	71.43	64.58	74.47	76.09	71.11	79.55	83.72	85.71	90.24	92.50	46.15	52.00	95.83
	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35
TR(%)	28.84	62.70	64.54	49.99	92.46	50.60	63.01	38.52	42.93	40.47	37.08	100.05	74.68	86.97	67.03	68.48	57.56
SR(%)	95.65	50.00	52.38	70.00	21.05	77.78	64.71	87.50	80.00	85.71	92.31	0.00	45.45	20.00	55.56	50.00	71.43

Table 5. TR and SR of each modification.

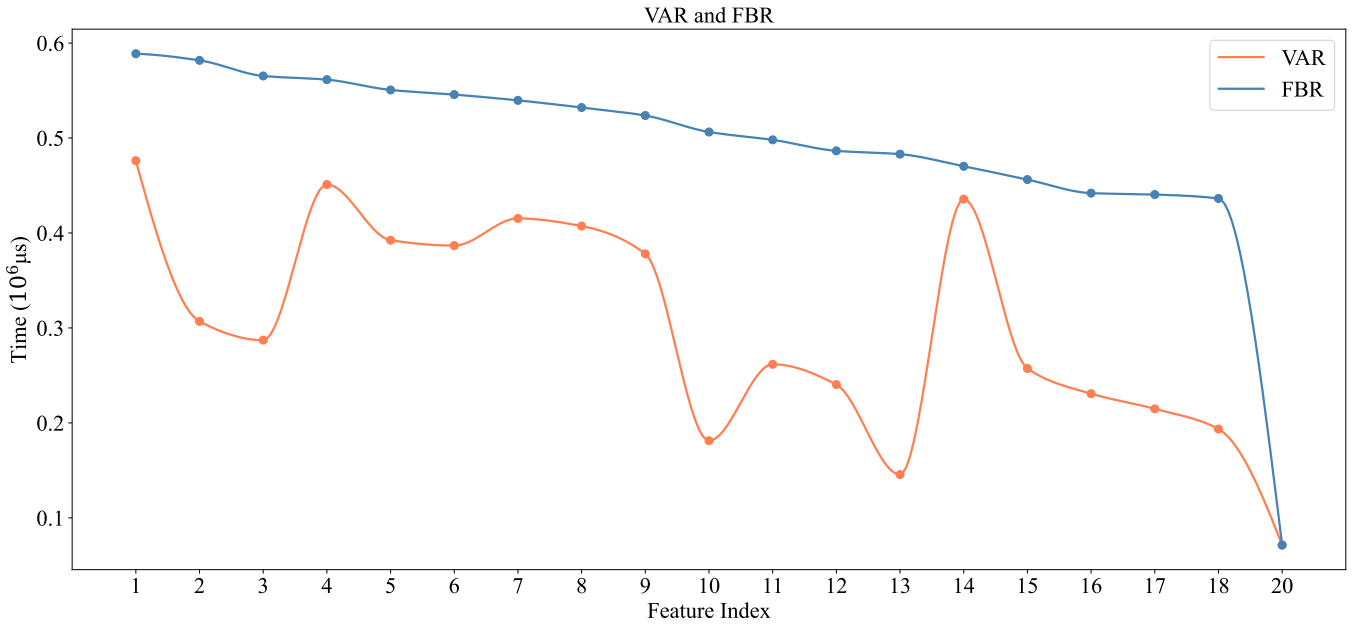


Fig. 15. Time comparison between VAR and FBR. The x-axis represents the feature index being modified. Red points and curve indicate the time consumed by FBR, while blue points and curve represent the time consumed by VAR. The points are connected using cubic Hermite splines.

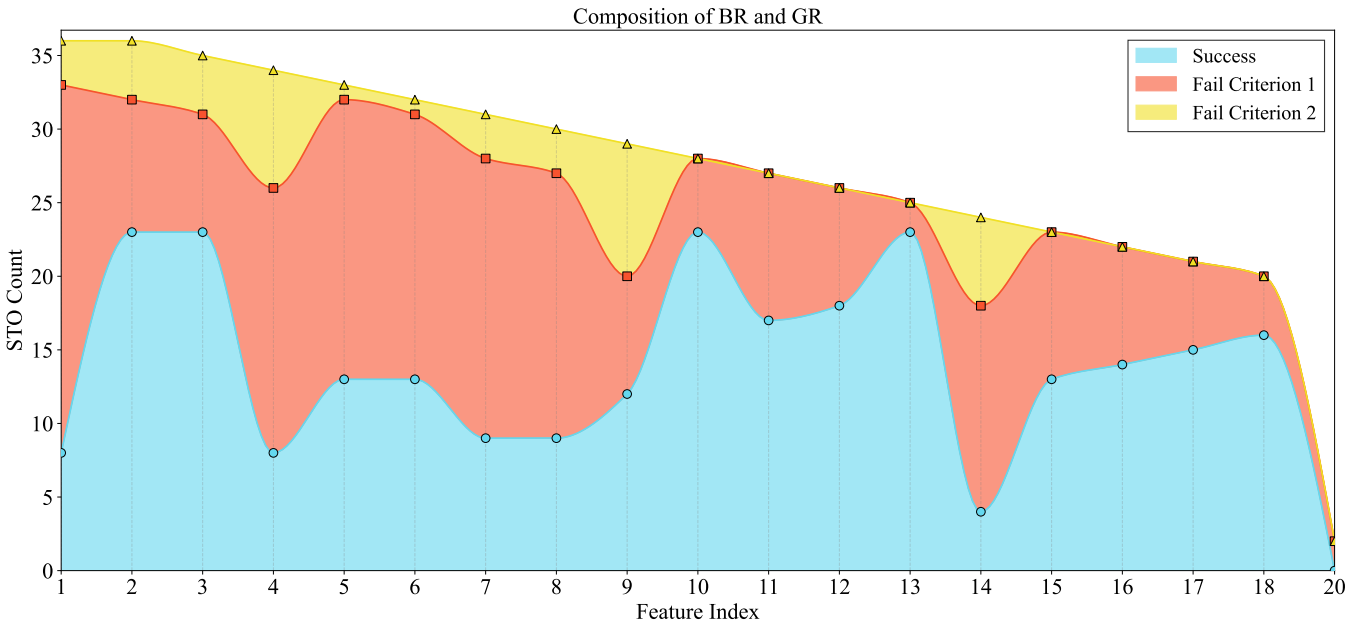


Fig. 16. STO composition breakdown for BR and GR. The x-axis denotes the modified feature index. The stacked vertical segments represent the composition of STOs: blue segments (from x-axis) show STOs that successfully executed GR(N_{GR}), red segments indicate STOs that failed criterion 1, and yellow segments indicate STOs that failed criterion 2. The points are actual measured data and are connected by cubic Hermite splines for each color.

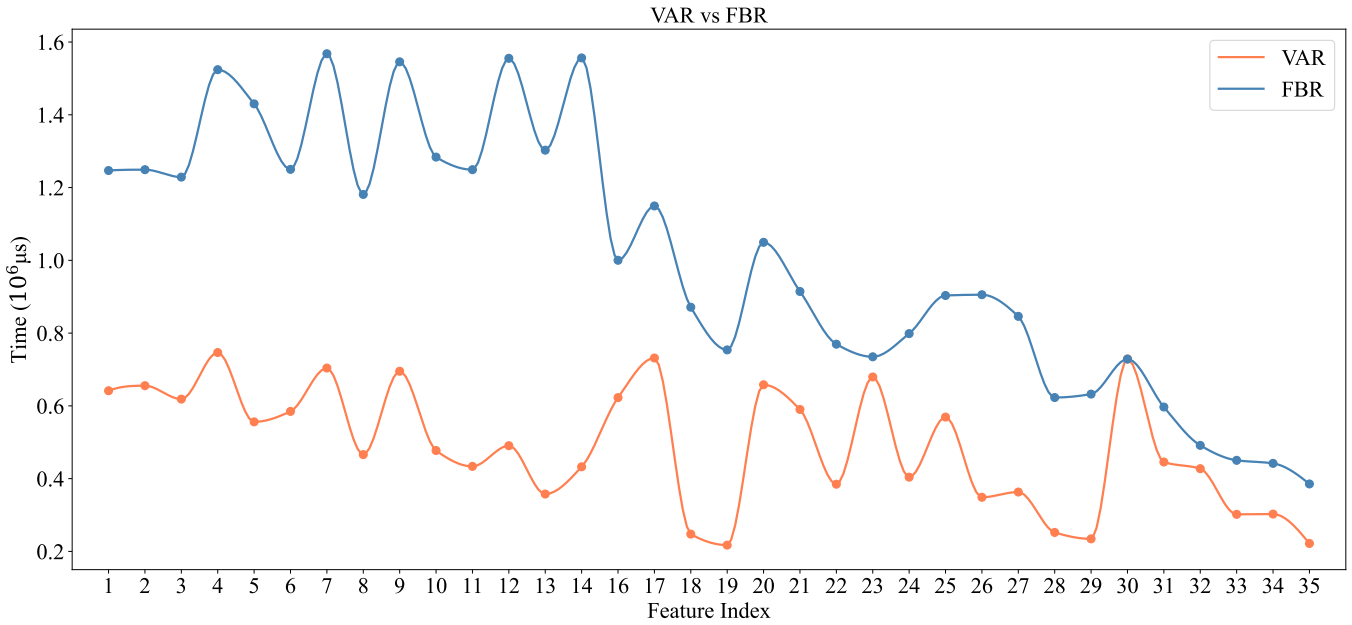


Fig. 17. Time comparison between VAR and FBR. The x-axis represents the feature index being modified. Red points and curve indicate the time consumed by FBR, while blue points and curve represent the time consumed by VAR. The points are connected using cubic Hermite splines.

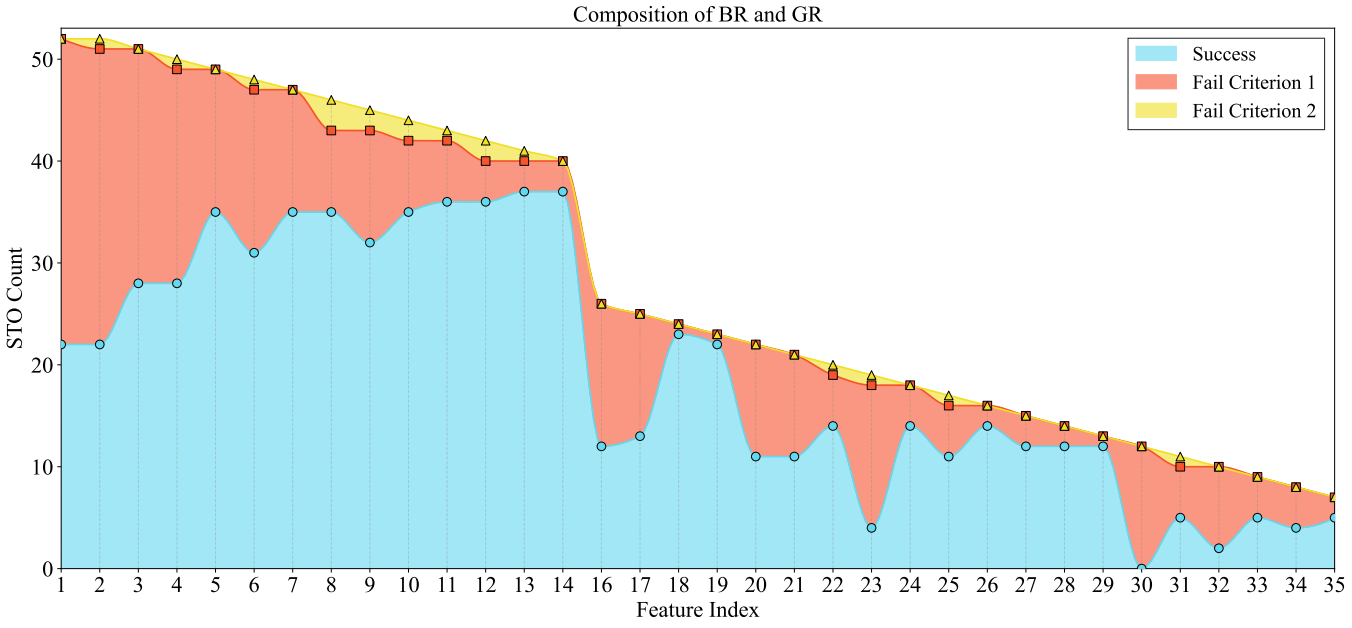
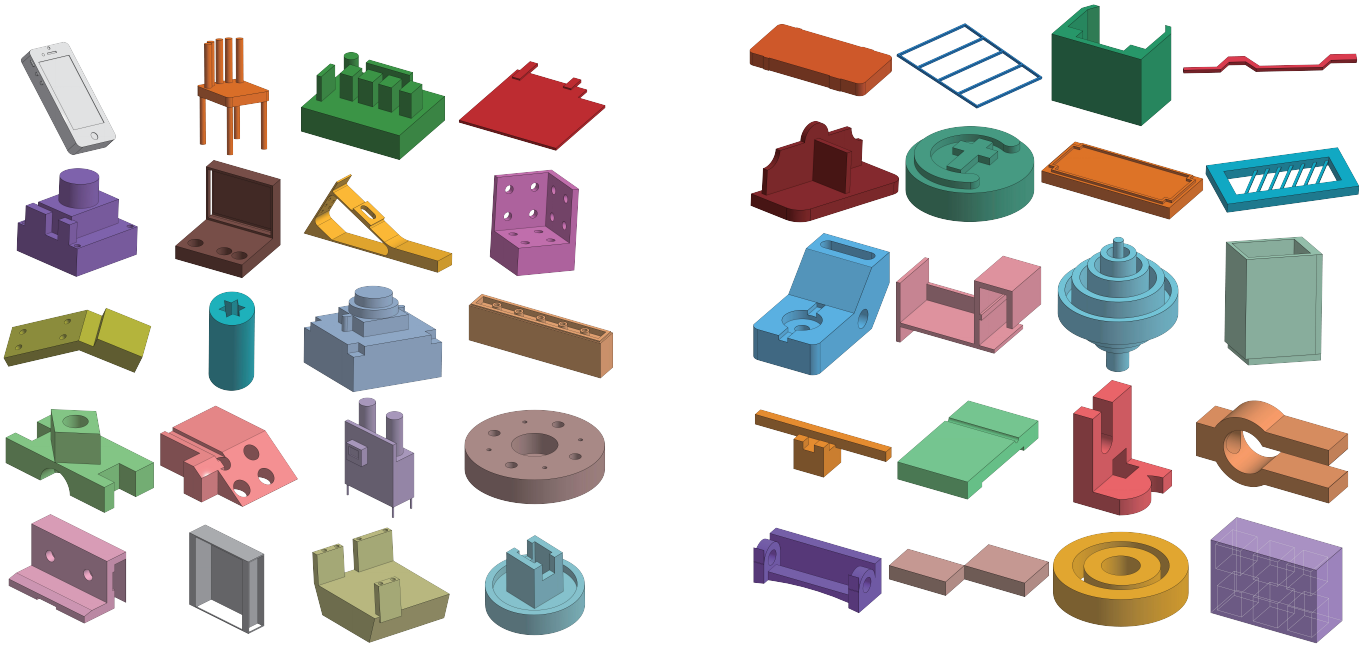


Fig. 18. STO composition breakdown for BR and GR. The x-axis denotes the modified feature index. The stacked vertical segments represent the composition of STOs: blue segments (from x-axis) show STOs that successfully executed GR(N_{GR}), red segments indicate STOs that failed criterion 1, and yellow segments indicate STOs that failed criterion 2. The points are actual measured data and are connected by cubic Hermite splines for each color.



(a) Models with locality.

(b) Models without locality.

Fig. 19. Model gallery : (a) are models in our dataset and (b) are models excluded from our dataset, these models can be roughly classified into five types, each one of them correspond to one line of models in (b). *Line 1:* The modeling approach is unreasonable. For example, these models could have been created by extruding a single sketch, but were instead constructed through multiple extrude operations of identical height. *Line 2:* Multiple features lie on the same plane and modifying any one of them causes downstream features to violate Criterion 1. *Line 3:* The model is global, meaning most features are strongly interdependent. In such cases, most features cannot successfully execute GR. *Line 4:* The model is overly simple, providing little value for verifying our algorithm. In these cases, FBR can be executed directly with negligible computational cost. *Line 5:* The models are invalid. For instance, the first two contain non-manifold boundaries, the third violates Assumption 2, and the last violates Assumption 1.

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